

Problem Set 3 – Suggested Answer

Exercises

1. $Z_N \xrightarrow{d} Z$ **implies** $Z_N = O_p(1)$.

We must show that Z_N is *bounded in probability*, i.e., for every $\epsilon > 0$ there exist finite Δ_ϵ and $N_\epsilon \in \mathbb{N}$ such that

$$\mathbb{P}(|Z_N| \geq \Delta_\epsilon) < \epsilon \quad \text{for all } N \geq N_\epsilon.$$

Step 1: Bound $\mathbb{P}(|Z| \geq c)$ using Chebyshev's inequality.

Since $EZ^2 < \infty$, Chebyshev's inequality gives, for any $c > 0$:

$$\mathbb{P}(|Z| \geq c) \leq \frac{EZ^2}{c^2}.$$

Given $\epsilon > 0$, choose c_ϵ large enough that $EZ^2/c_\epsilon^2 < \epsilon/2$, i.e., $c_\epsilon > \sqrt{2EZ^2/\epsilon}$. Also ensure that c_ϵ and $-c_\epsilon$ are continuity points of the CDF of Z (possible because a CDF has at most countably many discontinuities). Then

$$\mathbb{P}(|Z| \geq c_\epsilon) < \frac{\epsilon}{2}.$$

Step 2: Use convergence in distribution.

Since $Z_N \xrightarrow{d} Z$, by definition there exists $N_\epsilon \in \mathbb{N}$ such that for all $N \geq N_\epsilon$:

$$|\mathbb{P}(|Z_N| \geq c_\epsilon) - \mathbb{P}(|Z| \geq c_\epsilon)| < \frac{\epsilon}{2}.$$

(This applies because c_ϵ and $-c_\epsilon$ were chosen to be continuity points of the CDF of Z .)

Step 3: Combine.

For all $N \geq N_\epsilon$:

$$\mathbb{P}(|Z_N| \geq c_\epsilon) < \mathbb{P}(|Z| \geq c_\epsilon) + \frac{\epsilon}{2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Set $\Delta_\epsilon := c_\epsilon$. This establishes $Z_N = O_p(1)$. $\square$

2. Moment generating function of the normal distribution.

(a) **Deriving** $M_Y(t; (\mu, \sigma))$.

The MGF is $M_Y(t; (\mu, \sigma)) = E[e^{tY}]$:

$$M_Y(t; (\mu, \sigma)) = \int_{-\infty}^{\infty} e^{ty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) dy.$$

Combine the two exponential terms by completing the square. The combined exponent is

$$ty - \frac{(y-\mu)^2}{2\sigma^2} = -\frac{y^2 - 2(\mu + \sigma^2 t)y + \mu^2}{2\sigma^2}.$$

Complete the square in y :

$$y^2 - 2(\mu + \sigma^2 t)y + \mu^2 = (y - (\mu + \sigma^2 t))^2 - (\mu + \sigma^2 t)^2 + \mu^2.$$

Substituting back and factoring out the terms that do not depend on y :

$$M_Y(t; (\mu, \sigma)) = \exp\left(\frac{(\mu + \sigma^2 t)^2 - \mu^2}{2\sigma^2}\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - (\mu + \sigma^2 t))^2}{2\sigma^2}\right) dy.$$

The integral equals 1 because the integrand is the pdf of a $N(\mu + \sigma^2 t, \sigma^2)$ random variable. Simplifying the remaining exponent:

$$\frac{(\mu + \sigma^2 t)^2 - \mu^2}{2\sigma^2} = \frac{2\mu\sigma^2 t + \sigma^4 t^2}{2\sigma^2} = \mu t + \frac{\sigma^2 t^2}{2}.$$

Hence:

$$M_Y(t; (\mu, \sigma)) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

(b) **First two derivatives evaluated at** $t = 0$.

Using the chain rule on $M_Y(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$:

$$M_Y'(t; (\mu, \sigma)) = M_Y(t; (\mu, \sigma)) \cdot (\mu + \sigma^2 t).$$

Evaluating at $t = 0$:

$$M_Y'(0; (\mu, \sigma)) = e^0 \cdot \mu = \mu = E[Y],$$

confirming that the first derivative of the MGF at zero yields the first moment.

Differentiating once more (product rule):

$$\begin{aligned} M_Y''(t; (\mu, \sigma)) &= M_Y'(t) \cdot (\mu + \sigma^2 t) + M_Y(t) \cdot \sigma^2 \\ &= M_Y(t) \cdot (\mu + \sigma^2 t)^2 + M_Y(t) \cdot \sigma^2. \end{aligned}$$

Evaluating at $t = 0$:

$$M_Y''(0; (\mu, \sigma)) = \mu^2 + \sigma^2 = E[Y^2].$$

As a check: $\text{Var}(Y) = E[Y^2] - (E[Y])^2 = \sigma^2$, as expected.

(c) **Standard normal case.**

Setting $\mu = 0$ and $\sigma = 1$:

$$M_Y(t; (0, 1)) = e^{t^2/2}. \quad \square$$

This is the Lemma from the week 3 lecture notes.

3. **Properties of the projection and residual maker matrices.**

Recall: $P_X := X(X'X)^{-1}X'$ and $M_X := I_N - P_X$.

(i) $P_X Y = \hat{Y}$.

By the definition of OLS, $\hat{\beta}^{\text{OLS}} = (X'X)^{-1}X'Y$. Therefore:

$$P_X Y = X \underbrace{(X'X)^{-1}X'Y}_{=\hat{\beta}^{\text{OLS}}} = X\hat{\beta}^{\text{OLS}} = \hat{Y}. \quad \square$$

(ii) $M_X Y = \hat{u}$.

$$M_X Y = (I_N - P_X)Y = Y - P_X Y \stackrel{\text{(i)}}{=} Y - \hat{Y} = \hat{u}. \quad \square$$

(iii) $M_X u = \hat{u}$.

First observe that $M_X X = 0$:

$$M_X X = (I_N - P_X)X = X - X(X'X)^{-1}X'X = X - X = 0.$$

Then, using $u = Y - X\beta^*$:

$$M_X u = M_X(Y - X\beta^*) = M_X Y - \underbrace{M_X X}_{=0} \beta^* \stackrel{\text{(ii)}}{=} \hat{u}. \quad \square$$

(iv) **Symmetry:** $P_X = P_X'$ and $M_X = M_X'$.

Since $X'X$ is symmetric, $(X'X)^{-1}$ is also symmetric. Using $(AB)' = B'A'$:

$$P_X' = (X(X'X)^{-1}X')' = (X')'((X'X)^{-1})'X' = X(X'X)^{-1}X' = P_X. \quad \square$$

For M_X :

$$M_X' = (I_N - P_X)' = I_N - P_X' = I_N - P_X = M_X. \quad \square$$

(v) **Idempotency:** $P_X P_X = P_X$ and $M_X M_X = M_X$.

$$P_X P_X = X(X'X)^{-1}X' \cdot X(X'X)^{-1}X' = X(X'X)^{-1} \underbrace{(X'X)(X'X)^{-1}}_{=I_K} X' = X(X'X)^{-1}X' = P_X. \quad \square$$

Using the idempotency of P_X just proved:

$$\begin{aligned} M_X M_X &= (I_N - P_X)(I_N - P_X) \\ &= I_N - P_X - P_X + P_X P_X \\ &= I_N - 2P_X + P_X = I_N - P_X = M_X. \quad \square \end{aligned}$$

(vi) **tr** $P_X = K$ **and** **tr** $M_X = N - K$.

Key tool: the *cyclic property* of the trace, $\text{tr}(AB) = \text{tr}(BA)$. Apply it by cycling X' to the front:

$$\text{tr}(P_X) = \text{tr}(X(X'X)^{-1}X') = \text{tr}(X' \cdot X(X'X)^{-1}) = \text{tr}(X'X \cdot (X'X)^{-1}) = \text{tr}(I_K) = K.$$

By linearity of the trace and the result for P_X :

$$\text{tr}(M_X) = \text{tr}(I_N) - \text{tr}(P_X) = N - K. \quad \square$$

4. $\sum_{i=1}^N \hat{u}_i^2 / (N - K)$ **is an unbiased estimator of** σ_u^2 .

We assume the classical Gauss-Markov condition: $E[uu'|X] = \sigma_u^2 I_N$ (homoskedastic, serially uncorrelated errors).

Step 1: Write the sum as a quadratic form.

$$\sum_{i=1}^N \hat{u}_i^2 = \hat{u}'\hat{u}.$$

From Problem 3(iii), $\hat{u} = M_X u$, so using symmetry and idempotency of M_X :

$$\hat{u}'\hat{u} = (M_X u)'(M_X u) = u' M_X' M_X u = u' M_X u.$$

Step 2: Apply the trace trick.

Since $u' M_X u$ is a scalar it equals its own trace. Using $\text{tr}(AB) = \text{tr}(BA)$:

$$u' M_X u = \text{tr}(u' M_X u) = \text{tr}(M_X u u').$$

Step 3: Take conditional expectation given X .

Because M_X is a function of X alone, it is fixed under $E[\cdot | X]$:

$$E[\hat{u}'\hat{u} | X] = E[\text{tr}(M_X u u') | X] = \text{tr}(M_X E[uu' | X]) = \text{tr}(M_X \cdot \sigma_u^2 I_N) = \sigma_u^2 \text{tr}(M_X).$$

(Step 2 uses linearity of expectation and trace; step 3 uses the Gauss-Markov assumption; step 4 uses Problem 3(vi).) Since $\text{tr}(M_X) = N - K$:

$$E(\hat{u}^2 | X) = E\left[\frac{\sum_{i=1}^N \hat{u}_i^2}{N - K} \middle| X\right] = \frac{\sigma_u^2(N - K)}{N - K} = \sigma_u^2.$$

By the **law of iterated expectations**: $E\left[\sum_{i=1}^N \hat{u}_i^2 / (N - K)\right] = \sigma_u^2. \quad \square$

$$E(\hat{u}^2) = E(E(\hat{u}^2 | X)) = E(\sigma_u^2) = \sigma_u^2$$

5. Asymptotic distribution of $\sqrt{N}(\hat{\beta}^{\text{OLS}} - \beta^*)$ under homoskedasticity.

(a) Deriving the asymptotic distribution.

The OLS estimator satisfies

$$\hat{\beta}^{\text{OLS}} = (X'X)^{-1}X'Y = \beta^* + (X'X)^{-1}X'u,$$

so the centred and scaled estimator can be written as:

$$\sqrt{N}(\hat{\beta}^{\text{OLS}} - \beta^*) = \underbrace{\left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1}}_{A_N} \underbrace{\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i}_{B_N}.$$

Analyzing A_N . By the WLLN (iid observations, $E\|X_i X_i'\| < \infty$):

$$\frac{1}{N} \sum_{i=1}^N X_i X_i' \xrightarrow{p} E(X_i X_i') =: Q_{XX}.$$

Matrix inversion is continuous at non-singular matrices, so by the continuous mapping theorem:

$$A_N = \left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1} \xrightarrow{p} Q_{XX}^{-1}.$$

Analyzing B_N . Exogeneity gives $E(X_i u_i) = 0$, so B_N is centred. Under homoskedasticity, the variance of $X_i u_i$ simplifies:

$$\text{Var}(X_i u_i) = E(X_i u_i u_i X_i') = E[E(u_i^2 | X_i) X_i X_i'] = \sigma_u^2 E(X_i X_i') = \sigma_u^2 Q_{XX}.$$

(The second equality uses the law of iterated expectations; the third uses the homoskedasticity assumption $E(u_i^2 | X_i) = \sigma_u^2$.) By the CLT:

$$B_N = \frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} N(0, \sigma_u^2 Q_{XX}).$$

Combining via Slutsky's theorem. Since $A_N \xrightarrow{p} Q_{XX}^{-1}$ (a constant) and $B_N \xrightarrow{d} N(0, \sigma_u^2 Q_{XX})$, Slutsky gives $A_N B_N \xrightarrow{d} Q_{XX}^{-1} \cdot N(0, \sigma_u^2 Q_{XX})$. For a linear transformation $W \sim N(0, \Sigma)$ implies $CW \sim N(0, C\Sigma C')$, with $C = Q_{XX}^{-1}$ and $\Sigma = \sigma_u^2 Q_{XX}$:

$$Q_{XX}^{-1}(\sigma_u^2 Q_{XX})(Q_{XX}^{-1})' = \sigma_u^2 Q_{XX}^{-1} Q_{XX} Q_{XX}^{-1} = \sigma_u^2 Q_{XX}^{-1}.$$

Therefore:

$$\sqrt{N}(\hat{\beta}^{\text{OLS}} - \beta^*) \xrightarrow{d} N(0, \sigma_u^2 E(X_i X_i')^{-1}).$$

(b) A consistent estimator for the asymptotic variance.

The asymptotic variance is $\sigma_u^2 Q_{XX}^{-1}$. Replace each population quantity with a consistent sample analogue:

- $Q_{XX} = E(X_i X_i')$ by $\frac{1}{N} \sum_{i=1}^N X_i X_i'$,
- $\sigma_u^2 = E(u_i^2)$ by $s_u^2 := \frac{1}{N-K} \sum_{i=1}^N \hat{u}_i^2$ (unbiased by Problem 4).

A natural consistent estimator is therefore:

$$\hat{V}_N := s_u^2 \left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1}.$$

(c) **Proof that $\hat{V}_N$ is consistent.**

We must show $\hat{V}_N \xrightarrow{P} \sigma_u^2 Q_{XX}^{-1}$.

Part 1: $\left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1} = Q_{XX}^{-1} + \mathbf{o}_p(1)$.

By the WLLN:

$$\frac{1}{N} \sum_{i=1}^N X_i X_i' = Q_{XX} + \mathbf{o}_p(1).$$

Since inversion is continuous at the non-singular matrix Q_{XX} , Slutsky's theorem gives:

$$\left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1} = Q_{XX}^{-1} + \mathbf{o}_p(1).$$

Part 2: $s_u^2 = \sigma_u^2 + \mathbf{o}_p(1)$.

Write $\hat{u}_i = u_i - X_i'(\hat{\beta}^{\text{OLS}} - \beta^*)$ and expand:

$$\hat{u}_i^2 = u_i^2 - 2u_i X_i'(\hat{\beta}^{\text{OLS}} - \beta^*) + (\hat{\beta}^{\text{OLS}} - \beta^*)' X_i X_i' (\hat{\beta}^{\text{OLS}} - \beta^*).$$

Summing and dividing by $(N - K)$:

$$s_u^2 = \frac{1}{N-K} \sum_{i=1}^N u_i^2 - \frac{2}{N-K} \sum_{i=1}^N u_i X_i'(\hat{\beta}^{\text{OLS}} - \beta^*) + (\hat{\beta}^{\text{OLS}} - \beta^*)' \left(\frac{1}{N-K} \sum_{i=1}^N X_i X_i' \right) (\hat{\beta}^{\text{OLS}} - \beta^*).$$

Apply the WLLN and consistency of OLS to each term. Since $N/(N - K) \rightarrow 1$ (for fixed K):

$$\begin{aligned} \frac{1}{N-K} \sum_{i=1}^N u_i^2 &= E(u_i^2) + \mathbf{o}_p(1) = \sigma_u^2 + \mathbf{o}_p(1), \\ \frac{1}{N-K} \sum_{i=1}^N u_i X_i' &= E(u_i X_i') + \mathbf{o}_p(1) = \mathbf{o}_p(1) \quad (\text{since } E(u_i X_i') = 0), \\ \frac{1}{N-K} \sum_{i=1}^N X_i X_i' &= Q_{XX} + \mathbf{o}_p(1), \\ \hat{\beta}^{\text{OLS}} - \beta^* &= \mathbf{o}_p(1). \end{aligned}$$

Therefore:

$$s_u^2 = (\sigma_u^2 + \mathbf{o}_p(1)) - 2 \mathbf{o}_p(1) \cdot \mathbf{o}_p(1) + \mathbf{o}_p(1) \cdot \mathbf{O}_p(1) \cdot \mathbf{o}_p(1) = \sigma_u^2 + \mathbf{o}_p(1).$$

Part 3: Combine via Slutsky.

$$\begin{aligned}\hat{V}_N &= s_u^2 \cdot \left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1} \\ &= (\sigma_u^2 + \mathbf{o}_p(1))(Q_{XX}^{-1} + \mathbf{o}_p(1)) \\ &= \sigma_u^2 Q_{XX}^{-1} + \mathbf{o}_p(1) \xrightarrow{p} \sigma_u^2 Q_{XX}^{-1}. \quad \square\end{aligned}$$

(a)

$$\sqrt{N}(\hat{\beta}^{\text{OLS}} - \beta^*) = \left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i.$$

By WLLN

$$\frac{1}{N} \sum X_i X_i' = E(X_i X_i') + o_p(1)$$

By Slutsky's thm

$$\left(\frac{1}{N} \sum X_i X_i' \right)^{-1} = E(X_i X_i')^{-1} + o_p(1)$$

[supposing invertibility of $E(X_i X_i')$]

Applying CLT (noting $E(u_i X_i) = 0$)

$$\begin{aligned} \text{Var}(X_i u_i) &= E(u_i^2 X_i X_i') \\ &= E\left(E(u_i^2 X_i X_i' | X_i)\right) \\ &= E\left(X_i X_i' E(u_i^2 | X_i)\right) \\ &= \sigma_u^2 \cdot E(X_i X_i') \end{aligned}$$

$\Rightarrow$

$$\sqrt{N} \left(\frac{1}{N} \sum X_i u_i \right) \xrightarrow{d} N(0, \sigma_u^2 E(X_i X_i'))$$

Combining

$$\sqrt{n} (\hat{\beta} - \beta) \xrightarrow{d} N(0, Q)$$

$$\text{where } Q = \sigma_u^2 E(X_i X_i')^{-1}$$

(b) suggest consistent estimator:

$$s_u^2 := \frac{1}{n} \sum \hat{u}_i^2$$

where $\hat{u}$ are the residuals

$$\text{then } \hat{Q} := s_u^2 \left(\frac{1}{n} \sum X_i X_i' \right)^{-1}$$

(c)

We've already shown that

$$\left(\frac{1}{n} \sum X_i X_i' \right)^{-1} = E(X_i X_i')^{-1} + o_p(1)$$

What about s_u^2 ?

start with

$$\begin{aligned}\hat{u}_i &= y_i - x_i' \hat{\beta} \\ &= x_i' \beta^* + u_i - x_i' \hat{\beta} \\ &= u_i - x_i' (\hat{\beta} - \beta^*)\end{aligned}$$

Therefore

$$\begin{aligned}\hat{u}_i^2 &= u_i^2 - 2 u_i x_i' (\hat{\beta} - \beta^*) \\ &\quad + (\hat{\beta} - \beta^*)' x_i x_i' (\hat{\beta} - \beta^*)\end{aligned}$$

It follows that

$$\begin{aligned}s_u^2 &= \sum u_i^2 / N - 2 \sum u_i x_i' / N (\hat{\beta} - \beta^*) \\ &\quad + (\hat{\beta} - \beta^*)' \left(\frac{1}{N} \sum x_i x_i' \right) (\hat{\beta} - \beta^*)\end{aligned}$$

Considering each term (using WLLN)

$$\sum u_i^2 / N = E(u_i^2) + o_p(1) = \sigma_u^2 + o_p(1)$$

$$\bar{\sum} u_i X_i' / N = E(u_i X_i') + o_p(1) = o_p(1)$$

$$\frac{1}{N} \bar{\sum} X_i X_i' = E(X_i X_i') + o_p(1) = O_p(1)$$

$$\hat{\beta} - \beta^* = o_p(1)$$

Combining

$$s_u^2 = \sigma_u^2 + o_p(1) - \underbrace{2 \cdot o_p(1) \cdot o_p(1)}_{\text{blue}} + \underbrace{o_p(1) \cdot O_p(1) \cdot o_p(1)}_{\text{green}}$$

$$= \sigma_u^2 + o_p(1) - \underbrace{o_p(1)}_{\text{blue}} + \underbrace{o_p(1)}_{\text{green}}$$

$$= \sigma_u^2 + o_p(1)$$